

FACULTY OF ECONOMIC AND MANAGEMENT SCIENCES

DEPARTMENT OF ECONOMICS

ECONOMICS 114 A1

Date:	8 April 2025
Duration:	2 hours
Total Marks:	60
Examiners:	Dr C. Links (Convenor) Dr K. Malindi Dr M. Nchake Mr J. Pretorius Ms L. Dube Ms R. Petersen Ms J. Kazumunu
Internal Moderator	Dr M van Schoor

PLEASE NOTE: ONLY NON-PROGRAMMABLE CALCULATORS ARE ALLOWED.

ANSWER SECTION A's QUESTIONS ON THE SUPPLIED ANSWER SHEET. INDICATE YOUR CHOICE BY MAKING A CROSS NEXT TO THE NUMBER OF THE STATEMENT.

ALSO ANSWER SECTION B's QUESTIONS ON THE SUPPLIED ANSWER SHEET. INDICATE YOUR CHOICE BY MAKING A CROSS NEXT TO THE NUMBER OF THE STATEMENT.

ANSWER SECTION C's QUESTIONS IN THE ANSWER BOOK PROVIDED. START EVERY QUESTION ON A NEW PAGE. SHOW ALL CALCULATIONS.

Key Abbreviation	Meaning	Key Abbreviation	Meaning
AP	Average product	BL	Budget line
GDP	Gross domestic product	IC	Indifference curve
K	Capital	L	Labour
FF	Feasible frontier	MP	Marginal product
MRS	Marginal rate of substitution	MRT	Marginal rate of transformation
P	Price	PF	Production function
Q	Quantity	TP	Total product
w	Wage	Y	Income

SECTION A

TRUE OR FALSE STATEMENTS

[6 marks: 1 mark for a correct answer and $-\frac{1}{2}$ for an incorrect answer]

1. The capitalist revolution contributed to the widening income gap between nations.
 - a) True
 - b) False
2. The Malthusian model predicts that population increases lead to increases in income. But income cannot keep up with population growth and, therefore, the population will fall.
 - a) True
 - b) False
3. The MRT represents the slope of the indifference curve, $(\frac{p_y}{p_x})$ which tells us that rate at which the consumer must trade each good (x-axis) for a good on the y axis.
 - a) True
 - b) False
4. Before 1870, the substitution effect predominated over the income effect. However, in the 20th century, the income effect became more dominant, largely due to advancements in wage technology.
 - a) True
 - b) False
5. The prisoner's dilemma leads to a suboptimal Nash equilibrium where players can improve their outcome/ payoff when they can cooperate.
 - a) True
 - b) False
6. Nomsa signed a rental agreement stating that she would not have any pets in her apartment. A few months later, her landlord discovered that she had adopted a small service dog to assist with her anxiety disorder. The landlord issued an eviction notice, citing a breach of the lease agreement. Nomsa wrote a letter explaining that the dog is a medically prescribed service animal and requested an exemption from the pet policy. The housing tribunal reviewed her case and ruled that, while lease agreements must generally be upheld, disability accommodations must be considered under anti-discrimination laws. This represents an example of a procedural judgment of fairness on the part of the housing tribunal..
 - a) True
 - b) False

SECTION B

MULTIPLE CHOICE QUESTIONS

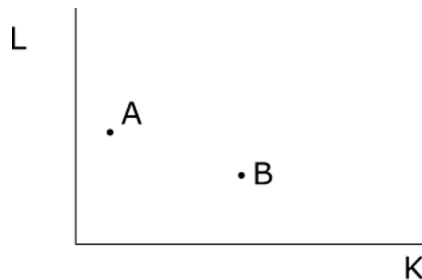
[24 marks: 1 mark for a correct answer and $-\frac{1}{4}$ for an incorrect answer]

7. The table below includes the current GDP values for South Africa for the period 2005 till 2010. Based on the provided GDP data, which statement is incorrect?

Year	2005	2006	2007	2008	2009	2010
GDP (R million)	1 639 254	1 839 401	2 109 501	2 369 063	2 507 677	2 748 008

Table 1: South Africa Annual Gross Domestic Product at Current Prices

- a) South Africa's economy performed better in 2007-2008 compared to 2009-2010.
 - b) The GDP of South Africa showed continuous growth from 2005 to 2010.
 - c) GDP alone does not account for the cost of environmental damage or income inequality.
 - d) A decrease in GDP always means a decline in the quality of life for all citizens.
8. In the graph below, K is the amount of capital required in production and L the amount of labour required in production, and **A** and **B** represent different technologies for producing 20 rugby balls.

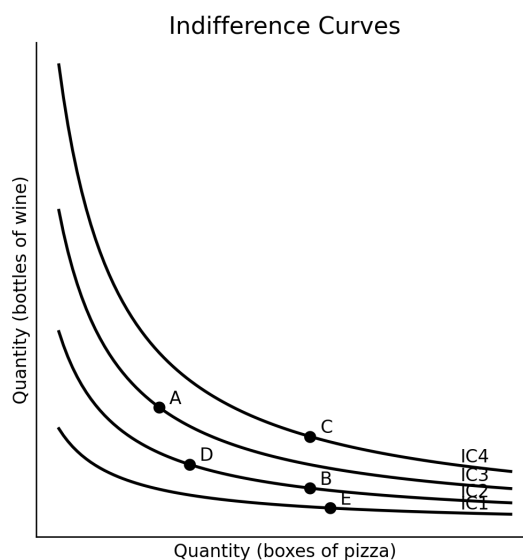


A is currently the cheapest technology. However, **B** could become the cheapest technology if

- a) the price of capital and labour both decrease by the same percentage.
- b) the price of capital and labour both increase, with a greater percentage increase for the price of labour.
- c) the price of capital and labour both decrease, with a greater percentage decrease for the price of labour.
- d) the price of labour increases while the price of capital remains constant.

9. Which of the following conditions are necessary for an economy to remain in the Malthusian trap?
- i) A concave production function
 - ii) Rising population in response to increasing wages
 - iii) Permanent improvements in technology that offsets the diminishing average product of labour.
 - iv) A convex production function
- a) i and ii
b) i, ii, and iii
c) ii, iii, and iv
d) None of the above.
10. The model of innovation argues that Britain escaped the Malthusian trap due to
- a) The cheaper relative cost of labour which created economic incentive for firms to switch to energy intensive technology.
 - b) The cheaper relative cost of coal which created economic incentive for firms to switch to energy intensive technology.
 - c) Due to better educational outcomes, Britain had a comparative advantage in innovation and saw sustained increases in productivity.
 - d) Britain had the cheapest coal.
11. If the MRS of a cup of coffee for bottle of juice is 2, this means:
- a) The consumer prefers coffee to juice.
 - b) The consumer is indifferent between a cup of coffee and a bottle of juice.
 - c) The consumer would be willing to give up 2 cups of coffee for 1 bottle of juice.
 - d) The consumer is willing to give up 1 bottle juice for 2 cups of coffee.

12. David chooses how many hours to spend studying for his A1 test. If the final grade he can achieve is given by a concave production function, this implies that the marginal product of each hour he uses to study _____ as the number of hours of study _____.
- decreases, increase
 - increases, decrease
 - decreases, decrease
 - increases, increase
13. A consumer who has a fixed income in Rand, consumes chocolate and noodles. Suppose that chocolate is a normal good, but noodles are an inferior good. When the price of chocolate increases, ceteris paribus, the income effect on chocolates is _____, while the substitution effect on noodles is _____.
- positive, negative
 - negative, negative
 - positive, positive
 - negative, positive
14. The figure below represents Sukho's indifference curves for pizza and wine.



Which of the following statements is CORRECT?

- Sukho prefers consumption bundle E above B.
- Sukho is indifferent between consumption bundle A and C.
- Sukho's most preferred consumption bundle is consumption bundle is C.
- The marginal rate of substitution is higher at B than at D.

Please use the information below to answer Questions 15 and 16

Aqueel and Curtly both like to speed, and like racing on their time off. But when they do speed, they have the potential of getting into a motor accident and potentially injuring innocent bystanders. They enjoy the thrill of racing and speeding, when one speeds, they win the race and the one that doesn't, gets caught by the traffic cop and is penalized.

		Player 2	
		Don't Speed	Speed
Player 1	Don't Speed	4, 4	1, 5
	Speed	5, 1	2, 2

15. The Nash Equilibrium of the game shown above is:
 - a) Don't speed, Don't speed
 - b) Don't speed, Speed
 - c) Speed, Don't speed
 - d) Speed, Speed
16. This game is an example of what type of game?
 - a) ultimatum game
 - b) invisible hand game
 - c) prisoner's dilemma
 - d) chicken game
17. In the economy of Namaland, there are two people, Elton and Merle, and two goods, copper and diamonds. Elton likes copper and dislikes diamonds, while Merle likes diamonds and dislikes copper. Due to a halt in mining activities, no copper is available this year, and only 12 diamonds are available due to minimal mining. What is the only Pareto efficient allocation in this economy?
 - a) Elton receives 0 diamonds, and Merle receives all 12 diamonds.
 - b) Elton receives 6 diamonds, and Merle receives 6 diamonds.
 - c) Elton receives 12 diamonds, and Merle receives 0 diamonds.
 - d) Any allocation is Pareto efficient as long as Merle gets more diamonds than Elton.
18. Suppose Warren is coerced into working for Ghislaine on her farm, producing white wine. Warren must give Samantha all the extra wine he produces beyond the amount needed for his survival. His production can be represented by a concave feasible frontier, and his needs by a convex biological survival constraint. Which ONE of the following statements is CORRECT?
 - a) Warren's production is always efficient.
 - b) Warren's production is always inefficient.
 - c) Warren's production is always efficient if he produces more than he needs.
 - d) Warren's production is always efficient if he produces less than he needs.

- a) If $MRT > MRS$, Ghislaine will choose Linda's working hours to be higher.
- b) If $MRT < MRS$, Ghislaine will choose a lower number of working hours for Linda .
- c) There will be only one Pareto efficient allocation of surplus between Warren and Ghislaine.
- d) A Pareto efficient allocation of white wine between Warren and Ghislaine does not necessarily mean that $MRT = MRS$.

SECTION C

DESCRIPTIVE QUESTIONS

Question 19 [3 marks]

Does Capitalism look the same for all countries? Justify your answer.

Question 20 [7 marks]

The following table shows the input costs of two different technologies in producing X amount of e-cigarettes (vapes). Assume that the production of e-cigarettes only requires two inputs to production: casings and wiring.

Technology	Casing (kg)	Wiring (m)
AirsPops	12	8
Vuse	8	12

Table 2: Comparison of Casing and Wiring for Different Technologies

20.1 Casings cost R10 per kilogram, and wiring costs R8 per meter

20.1.1 If casings are plotted on the vertical (y-) axis, derive the equation of the isocost lines for each of the two technologies.

[2]

20.1.2 What is the economic rent gained/lost by switching technology from AirsPops to Vuse? Is there economic incentive to switch from AirsPops to Vuse?

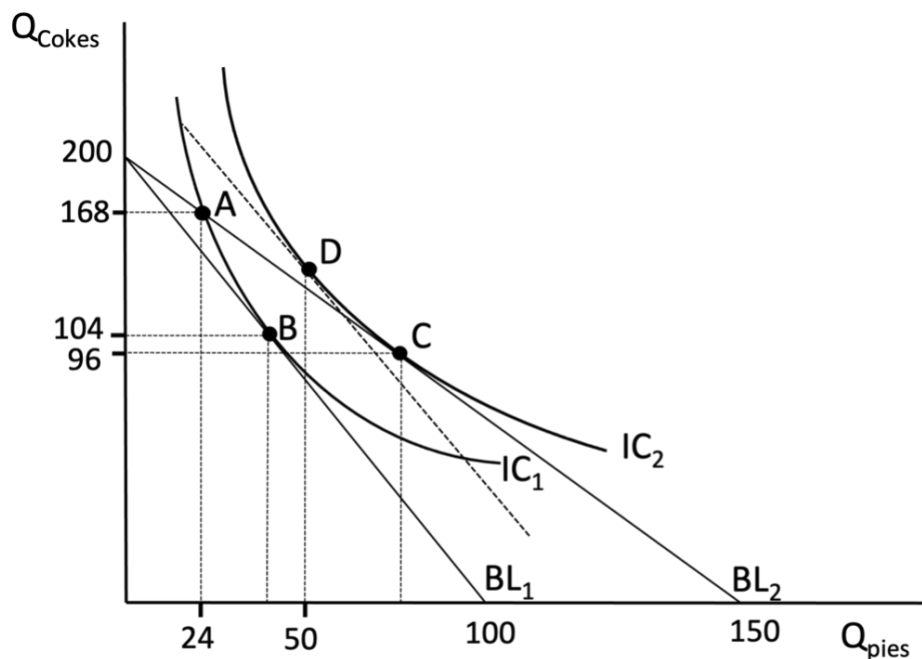
[2]

20.2 Suppose that the firm cannot determine the exact price of casing or wiring, but they do know the relative price of 1 metre of wiring is 1.8 kilograms of casing. What technology should the firm choose?

[3]

Question 21 [5 marks]

The figure below (not drawn to scale) indicates two budget lines, BL_1 and BL_2 , and two indifference curves of Linda, a consumer of Cokes and pies. Linda spends a total of R3 000 on these two goods. A decrease in the price of pies moves Linda's optimal consumption from point B to point C.



21.1 The effect of the decrease in the price of pies can be decomposed into an (i) income effect and (ii) substitution effect component. Explain with the aid of the graph below the decomposition of the overall effect.

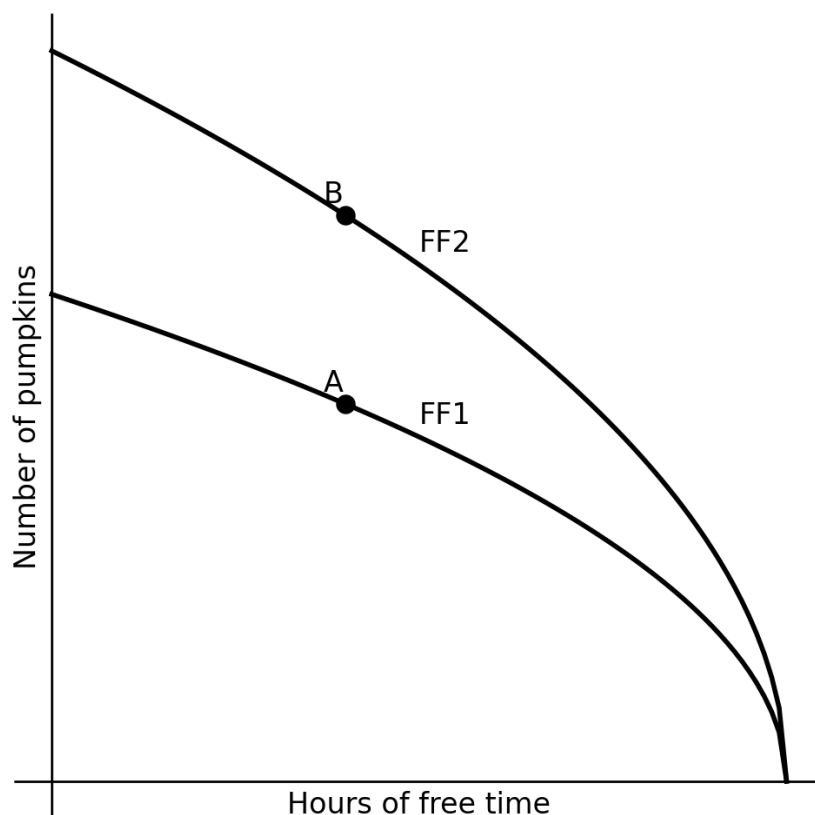
[3]

21.2 Does the income or substitution effect dominate for Coke? Explain your answer.

[2]

Question 22 [5 marks]

The graph below illustrates two feasible frontiers for Palesa, a pumpkin farmer.



22.1 Explain the economic forces that could cause the shift from FF_1 to FF_2

[3]

22.2 How does this shift affect the opportunity cost of free time?

[2]

Question 23 [5 marks]

Consider the game of Chicken represented in the table below:

		Player 2	
		Swerve	Don't Swerve
Player 1	Swerve	0, 0	-2, 0
	Don't Swerve	2, -2	-3, -3

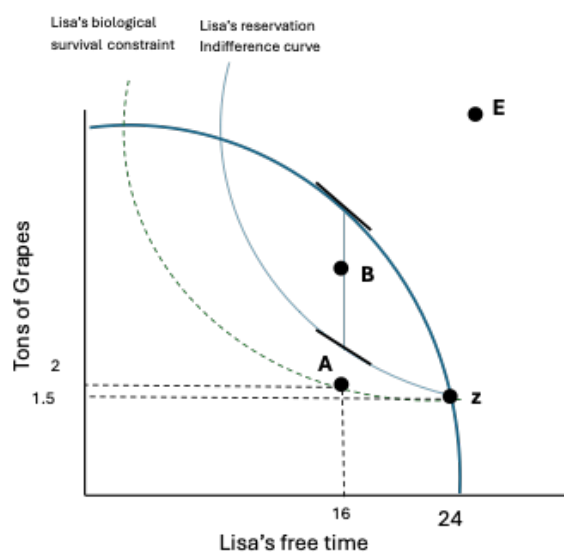
23.1 Find the Equilibrium solution(s) for the game.

[3]

23.2 Do any players have a dominant strategy? What is the dominant strategy if they do have one ?

[2]

Question 24 [5 marks]



24.1 In the diagram above, which combination of grapes per ton and free time (represented by points A, B, E, and Z) is unfeasible?

[1]

24.2 If Johan offers Lisa a wage of 1.5 tons of grapes, how many hours per day would she choose to work? Explain your reasoning.

[2]

24.3 At point A, if Johan offers Lisa a wage of 2 tons of grapes and expects her to work 8 hours per day, should she accept this offer? Explain your reasoning.

[2]